

Atmospheric Refraction

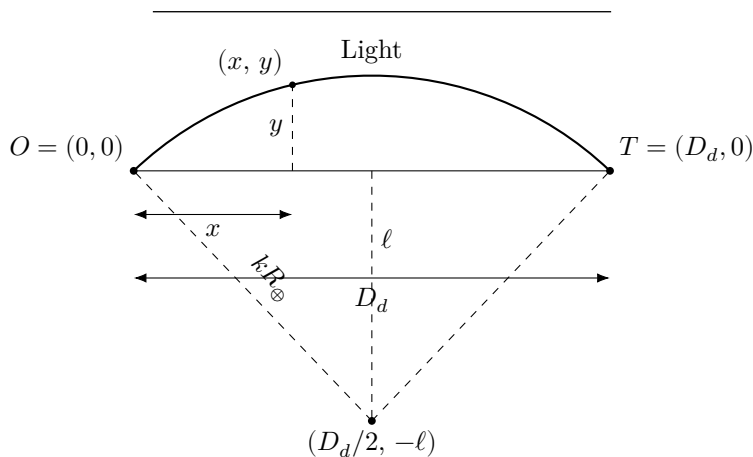
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Atmospheric refraction bends light, so a light ray is modelled as the arc of a circle of radius $R_{\text{light}} = k R_{\oplus}$. This document derives the height of that arc above the straight observer-target line (used as `light_heights` in `LineOfSight.process_full_line_of_sight` in `scripts/commons.py`).

Work in the rotated frame from `earth_curvature.md`: the observer is at $(0, 0)$ and the target is at $(D_d, 0)$, where D_d is the direct (straight-line) distance between them. Find y , the height of the light arc, in terms of x , k , D_d , and $R_{\oplus}$.

Notation

- y — height of the light ray above the straight observer-target line.
- x — horizontal distance along that straight line.
- k — light curvature factor, $R_{\text{light}} = k R_{\oplus}$. Larger k means straighter light.
- D_d — direct (straight-line) distance between observer and target.
- $R_{\oplus}$ — mean radius of the Earth at sea level.



The light ray as the upper arc of a circle of radius $kR_{\oplus}$ through O and T , centred a distance ℓ below the chord.

The light arc is a circle of radius $kR_{\oplus}$ passing through the observer and target. Its centre sits a distance $\ell = \sqrt{(kR_{\oplus})^2 - (D_d/2)^2}$ below the midpoint of the

chord, so the circle is

$$(kR_{\oplus})^2 = \left(x - \frac{D_d}{2}\right)^2 + (y + \ell)^2, \quad \ell = \sqrt{(kR_{\oplus})^2 - \left(\frac{D_d}{2}\right)^2}.$$

Expanding and simplifying:

$$\begin{aligned} (kR_{\oplus})^2 &= x^2 - D_d x + \left(\frac{D_d}{2}\right)^2 + y^2 + 2\ell y + (kR_{\oplus})^2 - \left(\frac{D_d}{2}\right)^2 \\ 0 &= y^2 + 2\ell y + x(x - D_d) \end{aligned}$$

Solving the quadratic for y (taking the upper arc):

$$y = \frac{-2\ell + \sqrt{(2\ell)^2 - 4x(x - D_d)}}{2} = \sqrt{\ell^2 - x(x - D_d)} - \ell$$

In `commons.py` this is written with $\gamma = \ell^2 = (kR_{\oplus})^2 - (D_d/2)^2$:

$$y(x) = \sqrt{\gamma + x(D_d - x)} - \sqrt{\gamma}.$$